

MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (90 minutes) December, 6th 2014

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. Rutgers Scientists have discovered how to generate a new atomic particle they have named the "SNOOKI"). SNOOKIs decay quickly after being generated having an exponential decay function. The probability a SNOOKI decays at " t " seconds after it is generated is te^{-at} for $a > 0$ unknown. It is impossible to observe the exact time of decay of a given particle, but Rutgers can determine if a SNOOKI has decayed prior to 10 seconds. For 25 SNOOKIs that were generated, 5 decayed in less than 10 seconds and 20 had not. Give a point estimate for a and a 95% confidence interval for a , the parameter of the decay function.
2. A researcher seeks to compare the sodium content in beef, chicken and turkey hotdogs. The researcher takes a random sample of 20 beef hotdogs, 20 poultry hotdogs, and 20 mixed meat hotdogs, and measures their sodium content. Assume sodium content is normally distributed. The following is the data on the means and standard deviations of sodium content:

	n	Mean	Standard Deviation
Beef	20	400	100
Poultry	20	480	90
Mixed Meat	20	410	85

- a) Test the null hypothesis that for sodium content, H_0 : Mean Beef = Mean Poultry = Mean Mixed meat, versus the alternative hypothesis Not H_0 .
- b) A school lunch program wants to compare the sodium content of a hotdog in two alternative diets:
Diet A: A Mixed Meat Hotdog
Diet B: half the time serve a Beef Hotdog, and half the time serve a Poultry Hotdog
Test the Null hypothesis that there is no difference in the average sodium content in the hotdog in the two diets, versus the alternative hypothesis that there is a difference.

3. A radiologist examines mammograms to detect breast cancer tumors. The goal is to detect tumors in their very early stages when they are very small. A company developed a computer program to be used by a medical technician to examine mammograms for the same purpose. A researcher had a sample of 100 mammograms that were determined (by subsequent biopsies) to have tiny tumors. The researcher had a radiologist review the 100 mammograms and for each state whether or not tumors were detected. The researcher also had a non-medical technician use the computer program to analyze the same 100 mammograms, and for each state whether or not tumors were detected. The null hypothesis is that the radiologist and computer program would equally often correctly detect tiny tumors. The research hypothesis is that the medical technician with the computer program would correctly detect tiny tumors more often than the radiologist. The researcher found that the radiologist detected tiny tumors in 65 of the mammograms, while the medical technician detected tumors in 80 of the mammograms. There were 5 mammograms where neither the radiologist, nor the medical technician with computer detected tumors. Carry out the test at the 0.05 level of significance. Show the details of your analysis including formula used and substitutions into formula.

4. Below are data on numbers of; AIDs Cases, AIDs Cases caused by PCP and AIDs Cases caused by KS for two American cities. (Note AIDS cannot be caused by both PCP and KS at the same time).

CITY	AIDS CASES		
	TOTAL	CAUSED BY PCP	CAUSED BY KS
Des Moines	200	30	30
El Paso	200	20	50

Let p_D be the portion of AIDS cases in Des Moines caused by PCP

p_E be the portion of AIDS cases in El Paso caused by PCP

k_D be the portion of AIDS cases in Des Moines caused by KS

k_E be the portion of AIDS cases in El Paso caused by KS

- Dr. LP believes that $p_E = p_D$. Use the data above to give a large sample test of a two sided alternative hypothesis against the appropriate null at $\alpha=0.05$
- Dr. PL believes $k_D = 2p_D$. Use the data above to give a large sample test of a two sided alternative hypothesis against the appropriate null at $\alpha=0.05$